TP04 - Docker and Docker Compose

Questions and Answers | English Version

Project: TalkingData Analysis Docker image: chebarim/talkingdata-analysis:v1

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Task 4 - Docker Compose

1. What problem does Docker Compose solve?

Docker Compose simplifies the configuration and management of applications with one or more containers. Services, networks, and volumes are defined in a single YAML file, so the environment can be started and stopped consistently with simple commands.

2. What is the difference between docker run and docker compose up?

docker run creates and starts a container using options supplied on the command line. docker compose up reads a Compose YAML file and creates or starts its configured services, networks, and volumes. In this TP, docker run launched three analysis containers separately, while docker compose up --build built and ran the analysis service.

3. How can you add a database service to Docker Compose?

Add another service, such as PostgreSQL, under services in compose.yaml. Configure its image, environment variables, and persistent volume. The analysis service can reach it using the database service name on the Compose network. The database integration is a proposed extension; it was not implemented in this TP. Production credentials should be stored securely.

Task 5 - Theoretical Questions

1. What is Docker, and why is it useful for Data Science?

Docker is a containerization platform that packages an application and its dependencies into a portable, isolated runtime. For Data Science, it improves reproducibility and consistency across machines. Here, Docker runs a Python 3.12 and Pandas application that analyzes a TalkingData CSV sample.

2. What is the difference between a Docker Image and a Docker Container?

A Docker image is a read-only template containing application code, dependencies, and runtime configuration. A container is a runnable instance created from an image. Several independent containers can use the same image. In this TP, three containers were created from chebarim/talkingdata-analysis:v1.

3. What is the purpose of FROM, WORKDIR, COPY, RUN, and CMD in a Dockerfile?

FROM selects the base image (python:3.12-slim). WORKDIR sets the working directory (/app). COPY transfers application and dependency files into the image. RUN executes commands during image building (such as installing Pandas). CMD defines the default command executed when a container starts (python -u app/app.py).